Derivatives and Fixed Income
PGE M1 Fall 2026

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Session 2: Time Value of Money, Forwards and Futures

BUSINESS SCHOOL

SKEMA

Introduction



About me

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Plan For Today



- Time Value of Money
- Forwards
- Futures

Future Value



Suppose the interest rate is 5%. What is the value of €100 in one year if you deposit it in a bank?

$$FV = 100 \times 1.05^1 = 105$$

For a single cash flow:
Future Value (FV) = $PV \times (1 + r)^T$

where

- $PV = \text{present value of the cash flow}$
- $r = \text{discount rate}$
- $T = \text{time}$

Monthly Compounding

Suppose you deposit €100 in a bank today at an annual rate of 5%. However, interest compounds **monthly**.

a. What is the value in your bank account after 1 year?

Formula:

$$\begin{aligned} FV &= PV \times (1 + r/m)^{mT} \\ &= 100 \times (1 + .05/12)^{12} \\ &= 105.1 \end{aligned}$$

Monthly Compounding

Suppose you deposit €100 in a bank today at an annual rate of 5%. However, interest compounds **monthly**.

b. What is the effective annual rate?

Formula:

$$\begin{aligned} EAR &= (1 + r/m)^{mT} - 1 \\ &= (1 + .05/12)^{12} - 1 = .051 \end{aligned}$$

Monthly Compounding



Suppose you deposit €100 in a bank today at an annual rate of 5%. However, interest compounds **monthly**.

b. What is the effective annual rate?

Quoting Rates

In general, the standard way of quoting rates is

$$r\% \text{ per annum, compounded } m \text{ times a year}$$

where compounded

- monthly $m = 12$
- quarterly: $m = 4$
- semi-annually: $m = 2$
- annually: $m = 1$

Quoting Rates



Suppose you deposit €100 in a bank today. What is value after 1 year if

a. The annual rate compounded semi-annually is 4%

Quoting Rates



Suppose you deposit €100 in a bank today. What is value after 1 year if

b. The semi-annual rate is 4%

Continuous Compounding

Suppose r is the annual rate, compounded rate continuously. Then,

$$FV = PV \times e^{rT}$$

Example: Suppose you deposit €100 in today. If the annual rate (compounded continuously) = continuously compounded annual rate is 5%. Then,

$$\begin{aligned} FV &= PV \times e^{rT} \\ &= 100 \times e^{.05} \\ &= 105.13 \end{aligned}$$

So the EAR is about 5.13%.

Continuous Compounding

Suppose r is the annual rate, compounded rate continuously. Then,

$$FV = PV \times e^{rT}$$

Exercise 4.3



An investor receives $110 in one year in return for an investment of $100 now. Calculate the percentage return per annum with:

a. Annual compounding

Exercise 4.3

An investor receives \$110 in one year in return for an investment of \$100 now. Calculate the percentage return per annum with:

b. Semiannual compounding

Start with

$$FV = PV \times (1 + r/m)^{mT}$$

$r = 9.76\%$

Exercise 4.3



An investor receives \$110 in one year in return for an investment of \$100 now. Calculate the percentage return per annum with:

c. Monthly compounding

$$
\begin{aligned}
FV &= PV \times (1 + r/m)^{mT} \\
110 &= 100 \times (1 + r/12)^{12} \\
\frac{110}{100} &= (1 + r/12)^{12} \\
1.1^{1/12} &= (1 + r/12) \\
r &= 12(1.1^{1/12} - 1) = 9.57\%
\end{aligned}
$$

Exercise 4.3



An investor receives \$110 in one year in return for an investment of \$100 now. Calculate the percentage return per annum with:

d. Continuous compounding

Start with

$$FV = PV \times e^{rT}$$

$r = \ln(1.1) = 9.53\%$

Exercise 4.24



An interest rate is quoted as 5% per annum with semiannual compounding. What is the equivalent rate with

a. Annual compounding

$r = 1.025^2 - 1 = 5.06\%$

Exercise 4.24



An interest rate is quoted as 5% per annum with semiannual compounding. What is the equivalent rate with

b. monthly compounding?

$$r = 12 \times \left[1.025^{1/6} - 1\right] = 4.96\%$$

Exercise 4.24



An interest rate is quoted as 5% per annum with semiannual compounding. What is the equivalent rate with

c. continuous compounding?

$r = 2\ln(1.025) = 4.95\%$

Exercise 4.9



A deposit account pays 4% per annum with continuous compounding, but interest is actually paid quarterly. How much interest will be paid each quarter on a $100 deposit?

Hint: First find the effective quarterly rate

Interest = $100 \times .01005 = 1$

Plan For Today



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Review: Forwards



Definition. A forward contract is an obligation to buy or sell an asset at a fixed time in the future for a specified price agreed upon today.

The fair price of a forward is given by

$$F = S_0 \times (1 + r)^T$$

where

- S_0 the value of the underlying asset today
- r interest rate per annum
- T time horizon

Review: Forwards



Definition. A forward contract is an obligation to buy or sell an asset at a fixed time in the future for a specified price agreed upon today.

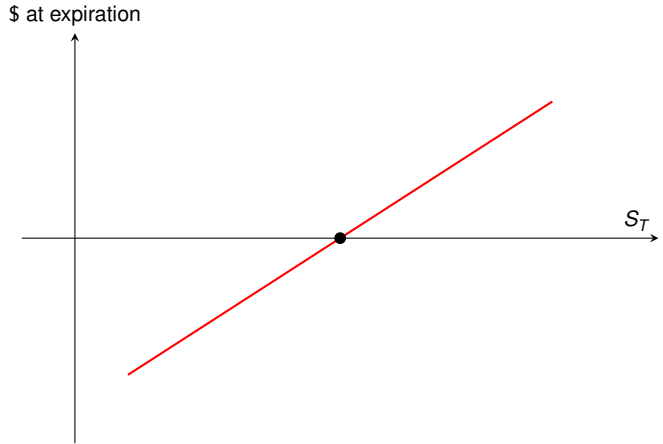
The fair price of a forward is given by

$$F = S_0 \times (1 + r)^T$$

This formula assumes

- ▷ No dividends paid by the underlying
- ▷ No storage costs
- ▷ No transaction fees

Payoff Diagram: Long Forward



Payoff Table

	Today	Payoff at T
Long Forward	0	$S_T - F$
Short Forward	0	$F - S_T$

Exercise 1.2



An investor enters into a **short** forward contract to sell 100 British pounds for U.S. dollars at an exchange rate of 1.3000 USD per pound. How much does the investor gain or lose if the exchange rate at the end of the contract is

a. 1.29

Exercise 1.2



An investor enters into a **short** forward contract to sell 100 British pounds for U.S. dollars at an exchange rate of 1.3000 USD per pound. How much does the investor gain or lose if the exchange rate at the end of the contract is

b. 1.32

Exercise 1.2



An investor enters into a **short** forward contract to sell 100 British pounds for U.S. dollars at an exchange rate of 1.3000 USD per pound. How much does the investor gain or lose if the exchange rate at the end of the contract is

b. 1.32

In general,

$$P\&L = \text{Quantity} \times (S_T - F) \times \text{Position}$$

where

$$\text{Position} = +1 \text{ if long forward}$$
$$\text{Position} = -1 \text{ if short forward}$$

Example



An investor enters into a long forward contract to buy 500 barrels of oil at €100 per barrel. What is the P&L if the market price of oil at expiry is €80?

Forward Price vs Value of the Contract



Suppose I entered a forward contract on an underlying asset with $S_0 = 100,\ r = 5\%$ with $T = 1$ year. The forward is priced at $F = 103$.

a. Is this forward fairly priced?

Forward Price vs Value of the Contract

Suppose I entered a forward contract on an underlying asset with $S_0 = 100,\ r = 5\%$ with $T = 1$ year. The forward is priced at $F = 103$.

b. How can I construct an arbitrage strategy?

	Today	Payoff at T
Forward		

Forward Price vs Value of the Contract

Suppose I entered a forward contract on an underlying asset with $S_0 = 100,\ r = 5\%$ with $T = 1$ year. The forward is priced at $F = 103$.

c. What is the value of my forward contract today?

$\rightarrow$The value of the contract is the amount at which I can sell my position to a buyer of the contract.

$PV = 1.9$

Forward Price vs Value of the Contract



Suppose I entered a forward contract on an underlying asset with $S_0 = 100,\ r = 5\%$ with $T = 1$ year. The forward is fairly priced at $F = 105$. Six months later, the value of the underlying increased to $S_{6/12} = 110$.

a. What is the value of my forward contract at $t = 6$ months?

Forward Price vs Value of the Contract

a. What is the value of my forward contract at $t = 6$ months?

$$f_t = \frac{F_t - F_0}{(1+r)^{T-t}}$$

where

- f : Value of the forward contract at date t
- F_t : Fair value of the forward at t
- F_0 : Forward price at date 0
- T: Expiration date
- t: Current date

Forward Price with Continuous Compounding



The fair value of a forward contract with continuous compounding is given by

$$F = S_0 e^{rT}$$

where

- S_0: the value of the underlying at date 0
- r: the is continuously compounded annual rate
- T: date to expiration (maturity)

Exercise 5.2



Suppose that you enter into a 6-month forward contract on a non-dividend-paying stock when the stock price is \$30 and the risk-free interest rate (with continuous compounding) is 5% per annum. What is the forward price?

$$F = S_0 e^{rT} = 30 \times e^{.05*.5} = 30.76$$

Storage Costs

Suppose I want to enter a long forward contract on gold. The current price of gold is €4000 per ounce. Let $r = 5\%$ and $T = 1$. The counterparty argues that I must compensate him/her for the cost of storing gold over the life of the contract at €4 per ounce. What is the fair value of the forward?

$$F\ (\text{with storage}) = (S_0 + U) \times (1 + r)^T$$

where

- U: the present **value** of storage costs

In general, storage costs for commodities are about x% of value

- Gold: .1%-.05%
- Oil: 1%-3%
- Wheat/corn: 5%-10%

Leasing

Suppose I want to enter a long forward contract on gold. The current price of gold is €4000 per ounce. Let $r = 5\%$ and $T = 1$. The counterparty argues that I must compensate him/her for the opportunity cost of leasing/lending gold at rate $q = 2.5\%$.

$$F \text{ (with leasing)} = S_0 \times \left(\frac{1+r}{1+q}\right)^T$$

where

- q: the leasing rate

Continuous Versions of Storage and Leasing

Fair value of a forward

$$F = S_0 e^{rT}$$

with storage

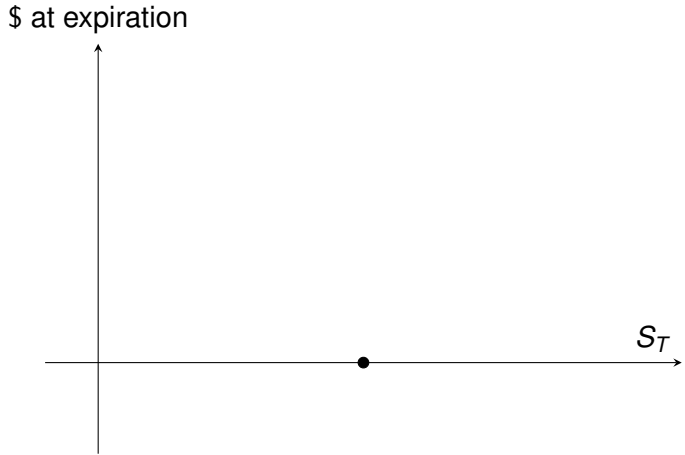
$$F = (S_0 + U)e^{rT}$$

with leasing

$$F = S_0 e^{(r-q)T}$$

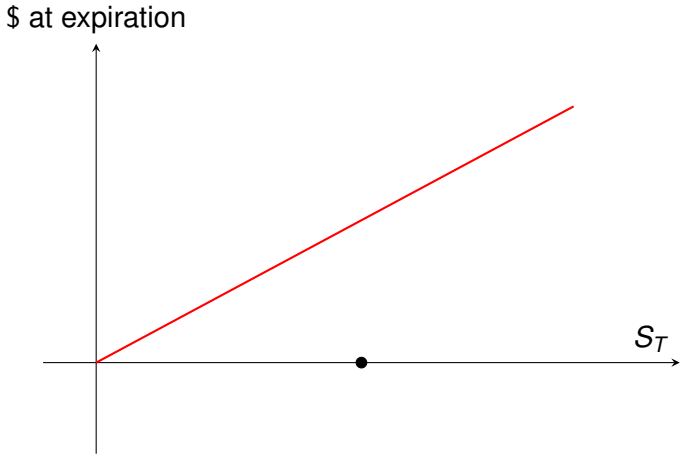
Payoff Diagram: Long Stock

Suppose I currently own a stock with value S_0.



Payoff Diagram: Long Stock

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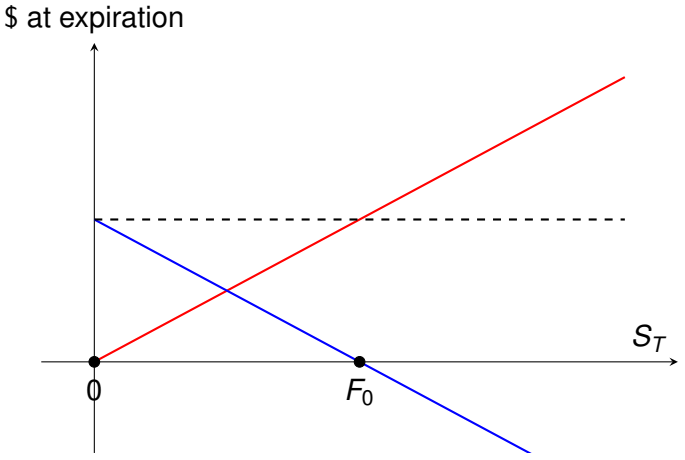


Hedging with Forwards



Suppose I currently own a stock with value S_0. I would like to hedge against a *decrease* in the stock price. As a result, I enter a short forward position with delivery price F. What is my payoff at T given r?

Long Stock + Short Forward



Exercise 1.31 (DIY)



The price of gold is currently $1,200 per ounce. The forward price for delivery in 1 year is $1,300 per ounce. An arbitrageur can borrow money at 3% per annum. What should the arbitrageur do? Assume that the cost of storing gold is zero and that gold provides no income.

Exercise 5.7 (DIY)

A 1-year long forward contract on a non-dividend-paying stock is entered into when the stock price is $40 and the risk-free rate of interest is 5% per annum with continuous compounding. (a) What are the forward price and the initial value of the forward contract? (b) Six months later, the price of the stock is $45 and the risk-free interest rate is still 5%. What are the forward price and the value of the forward contract?

Plan For Today



- Time Value of Money
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Futures

Problem with forwards: too much counterparty risk.

Futures are forward contracts that are traded on centralized exchanges rather than 1-to-1 via OTC markets.

In futures markets,

- Contract sizes are standardized
  - For example, crude oil futures are traded in increments of 1,000 barrels
- Profits and losses are settled daily (marked-to-market)

Futures

The average total daily dollar value of trading is

- Futures markets: $\sim$ 2 trillion
- Stock markets: $\sim$ 500 billion

How Trading in Futures Works



- Open a position
  Post an **initial margin**: a deposit to the exchange (normally around 5-15% of the contract value)
- **Marked-to-market**
  At the end of the trading day, the balance of your margin account updates given the value of the underlying asset
- Maintenance margin
  The value of my margin account must stay above the level of my maintenance margin.
- If the value of my margin account $<$ the maintenance margin, I receive a **margin call** in which I must top-up to the initial margin

Trading on Margin

Suppose I want to trade WTI crude oil futures.

- Position: Long 1 contract
- Contract size: 1000 barrels
- Current price: €100/barrel
- Total contract value: €100 $\times$ 1000 = 100.000
- Initial margin: 6.000 (6%)
- Maintenance margin: 4.500

Trading on Margins

Day	Settlement Price	Daily Change	Daily P&L	Margin Balance
0	€100,00	—	—	€6.000
1	€101,50	+€1,50	+€1.500	€7.500
2	€100,80	-€0,70	-€700	€6.800
3	€98,30	-€2,50	-€2.500	€4.300
		Margin Call		+ €1.700
4	€99,50	+€1,20	+€1.200	€7.200

Black Monday

October 19, 1987.
S&P 500 drops 20.4% on a single day.

Timeline:

i. Investor worries about higher US interest rates $\rightarrow$ stock prices fall

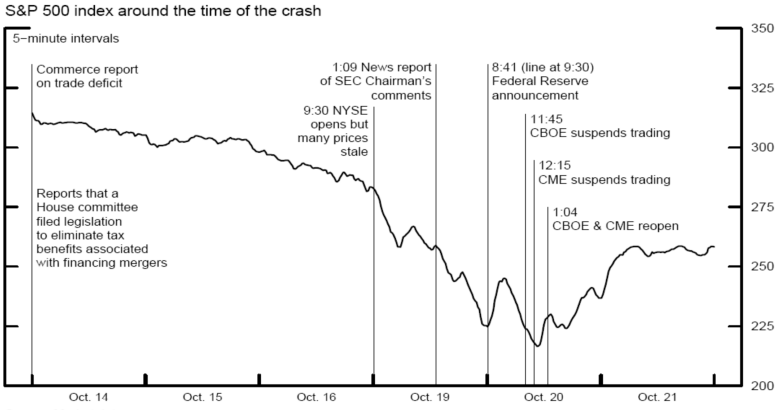
ii. Pension funds with computer programs start shorting S&P futures to protect against future losses.

iii. All pension funds are shorting $\rightarrow$ not enough buyers to take the opposite position $\Rightarrow$ market value of futures deviates from fair value

iv. Previous investors who took the long position in S&P futures face margin calls due to a sudden price drop

v. Investors sell their stocks in S&P to cover the margin call $\rightarrow$ lead to further drops in the S&P $\rightarrow$ more margin calls

Black Monday



Crises due to margin calls are pretty frequent



- 2022 UK Gilt Crisis
- 2020 COVID Crash
- 2018 Volmageddon
- 2010 Flash Crash

Trading on Margin

Recall

- Position: Long 1 contract
- Contract size: 1.000 barrels
- Current price: €100/barrel
- Total contract value: €100$\times$ 1000 = 100.000
- Initial margin: 6.000 (6%)
- Maintenance margin: 4.500

By how much does the price of oil fall by in order to trigger a margin call?

Margin Call

A margin call is triggered against the trader when the price of the underlying moves by more than:

$$\Delta = \frac{\text{Initial} - \text{Maintenance}}{\text{Quantity}}$$

In our case,

$$\Delta = \frac{\text{Initial} - \text{Maintenance}}{\text{Quantity}} = \frac{6.000 - 4.500}{1.000} = €1,5/\text{barrel}$$

Exercise 2.1

Suppose that you enter into a short futures contract to sell July silver for $17.20 per ounce. The size of the contract is 5,000 ounces. The initial margin is $4,000, and the maintenance margin is $3,000. What change in the futures price will lead to a margin call? What happens if you do not meet the margin call?

$\Delta = 0.2$ per ounce

Exercise 2.19

A cattle farmer expects to have 120,000 pounds of live cattle to sell in 3 months. The live cattle futures contract traded by the CME Group is for the delivery of 40,000 pounds of cattle. How can the farmer use the contract for hedging? From the farmer's viewpoint, what are the pros and cons of hedging?

Short 3 contracts

Exercise 2.7

A trader buys two July futures contracts on frozen orange juice concentrate. Each contract is for the delivery of 15,000 pounds. The current futures price is 160 cents per pound, the initial margin is $6,000 per contract, and the maintenance margin is $4,500 per contract. What price change would lead to a margin call? Under what circumstances could $2,000 be withdrawn from the margin account?

If prices increase by more than .067 cents per pound